ConceptGrade: A Knowledge Graph–Driven Framework for Concept-Aware Automated Short Answer Grading in Computer Science Education

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Abstract*—How well does a hybrid Knowledge-Graph + Large Language Model (KG-LLM) system actually work for Automated Short Answer Grading (ASAG) in Computer Science education, and where does it stop working? We present ConceptGrade, a five-layer framework that converts each student answer into a *concept graph* and compares it against an expert-built Data Structures knowledge graph (101 concepts, 138 typed relationships). We test it on $n = 2{,}276$ graded answers across 213 questions from three independent datasets (Mohler 2011, DigiKlausur, Kaggle ASAG), against a zero-shot LLM baseline that uses the identical underlying model. On its best dataset, ConceptGrade lowers error by 8.2% (Mohler 2011), though the gain is only marginally significant once question non-independence is accounted for. DigiKlausur shows a smaller but real gain. Kaggle ASAG shows none: its vocabulary falls entirely outside the knowledge graph. A real-data ablation tells us why the headline number is so modest in the first place – the deterministic knowledge-graph score, on its own, is 86.9% *worse* than the zero-shot baseline. Almost all of ConceptGrade's net gain turns out to come from its LLM verifier stage, not the knowledge-graph grounding the architecture is named for. Most of the raw-error advantage traces to fixing a simple scale bias rather than better ranking, and a promising-looking self-consistency ensembling gain mostly disappears once the baseline receives the same number of model calls. We report all of this as evidence of where KG-grounded ASAG is fragile and where its limits lie, not as proof that the approach generally works.

***Index Terms*—automated short answer grading, knowledge graph, concept extraction, computer science education, large language models, Bloom's taxonomy, NLP**

I. Introduction

This paper reports results on the Mohler et al. (2011) benchmark (46 questions, 1,262 responses, sourced from the public `nkazi/MohlerASAG` mirror, CC-BY-4.0) throughout, including a cross-dataset meta-analysis. The full six-condition component ablation from our original design requires additional pipeline runs and is reported as future work; in its place, a 3-condition ablation (C_LLM, pre-verifier `kg_score`, C5_fix) was reconstructed from already-cached components at zero new API cost (§VI) and found the KG-grounded score, taken alone, to be substantially *worse* than the zero-shot baseline – essentially all of ConceptGrade's small net gain comes from the verifier stage, not the KG comparison. The frozen Sentence-BERT baselines are reported in §V-A.

Scope and framing. This paper is not claiming that multi-layer knowledge-graph grounding is a general solution for short-answer grading. Our evidence points the other way: at best, the architecture gives a small, statistically fragile improvement in its single most favourable setting, and does nothing or actively hurts in nearby domains. Our claims rest on a pooled $n = 2{,}276$ paired predictions across three independent ASAG datasets and 213 questions (Section V-C), summarised here in five points:

1) On the KG-aligned Mohler 2011 sample ($n = 1{,}262$, 46 questions, `nkazi/MohlerASAG` mirror), ConceptGrade beats the identical zero-shot LLM by 8.2% paired MAE ($d_z = -0.154$, $p < 0.0001$ response-level), weakening to only marginal significance once grouped by question ($p = 0.111$ two-tailed; 17/46 LOOCV folds significant; wins only 27/46 questions). Pearson correlation is actually *worse* for ConceptGrade ($r = 0.784$ vs. 0.790).
2) A 3-condition ablation (§VI) finds the KG-grounded score alone far *worse* than the zero-shot baseline (-86.9% MAE) — almost all of ConceptGrade's small net gain comes from the verifier stage, not the knowledge graph. The full six-condition ablation is reported as future work, and non-LLM neural baselines are reported in §V-A.
3) Pooling across all three datasets gives a fixed-effects $d_z = -0.105$ ($p < 0.0001$), but the datasets disagree substantially ($I^2 = 73.2\%$), and the more conservative random-effects 95% CI $[-0.169, +0.002]$ essentially touches zero — the pipeline does not reliably generalise across datasets.
4) On Kaggle ASAG, concept extraction returned an empty matched-concept set for 368 of 368 unique answers (473 of 473 before deduplication), exactly the vocabulary-mismatch outcome we would expect given the KG's

domain, explaining the null result there.

5) Self-consistency ensembling (§V-E: 7 independent gradings at temperature 0.7, averaged, fixed before touching the evaluation data) improves response-level MAE over a single-call baseline on Mohler and DigiKlausur ($p < 0.0001$), but against an equally-resourced 7× baseline the strong question-level robustness we first reported (50/50, 17/17 folds) mostly disappears (Mohler 0/46; DigiKlausur 5/17/2/17). The response-level MAE gap survives on both (+7.0%/+6.4%, $p < 0.001$), but DigiKlausur's correlation advantage flips (0.727 vs. 0.735). No benefit on Kaggle ASAG, consistent with point 4.

Comparisons against frontier LLMs, retrieval-augmented baselines, and an independently built knowledge graph are left for future work (§VII-B). This paper's purpose is to document *when and why* a KG-LLM pipeline like this fails to add up to more than its parts in the ASAG setting. Readers looking for a triumphant new method should read that framing as intentional, not as an oversight.

Providing timely, consistent, and educationally meaningful feedback on student short answers is a critical pedagogical task in CS courses. Instructors routinely evaluate hundreds of written responses per assignment, each requiring domain-specific judgment about conceptual correctness, completeness, and depth; manual grading at this scale is labor-intensive, inconsistent across raters, and slow enough that students can no longer act on the feedback.

Automated Short Answer Grading (ASAG) systems aim to replicate expert grading behavior, but the CS domain presents its own set of problems. A student can correctly define a stack without ever using the word "LIFO," which is enough to trip up bag-of-words approaches. Two answers can use entirely different vocabulary and still reflect equivalent understanding. A technically correct sentence buried in a response full of misconceptions shouldn't earn full credit just for being technically correct. And the *structure* of what a student knows matters: whether concepts are merely listed or actually integrated with each other reflects qualitatively different levels of understanding.

Contemporary ASAG approaches fall into three broad families. *Lexical approaches* [3] use TF-IDF or n-gram overlap: fast, but blind to semantic equivalence. *Semantic/embedding approaches* [5], [7] use sentence-level alignment or pre-trained transformers such as BERT [6]. Sultan et al. (2016) achieved $r = 0.592$ without neural representations, and Sung et al. (2019) [7] report gains on the SemEval-2013 benchmark, though on a different metric (macro-F1) that we can't line up against a comparable Mohler r figure. Either way, these systems lack explicit domain knowledge and need labeled fine-tuning data we don't have for our KG-aligned subset. *LLM zero-shot approaches* prompt large models like GPT-4, Gemini, or Llama directly [15], [16], and they are surprisingly capable – our own zero-shot baseline reaches $r = 0.790$ on the real Mohler sample (§V) – but they stay opaque about what actually happened: no picture of what the student understood, noticeable run-to-run variance, nothing that names a missing concept. ConceptGrade instead uses the LLM as one structured component (concept extraction), not as an end-to-end black box.

We deliberately compare against *the same* `gemini-2.5-flash` *model* used for the zero-shot baseline, rather than a stronger frontier model, so that model capability stays out of the comparison: `C_LLM` and `C5_fix` share the model, the temperature ($T = 0$, Gemini batch API), and the rubric. The only real differences are that `C5_fix`'s prompt receives KG-grounding evidence and its synthesis weights were tuned on a development split, while `C_LLM` got no equivalent tuning (§VII-B). That means the reported gain reflects the KG-grounding architecture *and* its tuning budget together, not KG-grounding on its own. Frontier and retrieval-augmented baselines are left to future work on purpose, not because we forgot about them.

None of these approaches answer the questions an instructor actually needs answered: which concepts did the student demonstrate, which are missing from their response, and how deeply integrated is their understanding?

This paper presents **ConceptGrade**, a knowledge-graph-driven framework built around one idea: grading a CS short answer can be treated as a *knowledge graph comparison problem*. Transform the student's response into a concept graph, measure how well it matches an expert-curated domain KG, and the score follows – along with fine-grained diagnostic feedback that a bare number never could. We built this architecture to test that idea rigorously rather than assume it works, and testing it rigorously – including the ways it falls short – is this paper's central contribution.

Contributions. This paper makes the following contributions:

1) A five-layer ASAG architecture combining concept extraction, KG comparison, cognitive depth assessment, misconception detection, and score synthesis.
2) An expert Data Structures domain KG with 101 concepts and 138 typed relationships, manually curated following CS education ontology principles.
3) A hybrid LLM + rule-based concept extraction pipeline that produces structured `StudentConceptGraph` objects from free-text answers.
4) A KG comparison module computing coverage, integration, and accuracy scores via structural graph matching.
5) A rigorous, multi-dataset empirical characterization of *when and why* the architecture's benefit appears, weakens, and disappears – including a real-data ablation showing the KG-grounding component is a net negative in isolation, and a cross-dataset boundary analysis tying the effect's size to vocabulary distance from the KG (§V-C).
6) A diagnostic failure-mode analysis of two live scoring defects, including five pre-registered repair attempts, none of which improved performance under the evaluated criteria – reported as evidence against the repair mechanism tried, not as a solved bug list (§VI-B).

Every quantitative claim in this manuscript is cross-checked against the codebase and cached evaluation results (`verify_all_paper_claims.py`); only currently-verified numbers and descriptions are reported below.

II. Related Work

A. Automated Short Answer Grading

ASAG research spans three decades, from hand-crafted rubrics [3], [4] ($r = 0.493/0.518$ on the CS benchmark) through sentence-level semantic similarity [5] ($r = 0.592$ on a held-out split not directly comparable to our real, KG-aligned $n = 1{,}262$/46-question subset, §V) to transformer fine-tuning on SemEval [1], [7]. Despite these advances, few prior ASAG systems for CS build a domain knowledge graph directly from the student's free-text response and use graph-structural comparison as the primary grading signal. Most still treat the task as sentence-pair similarity regression, which conflates syntactic fluency with conceptual accuracy and can't tell you which specific concepts are present, absent, or incorrectly related.

B. Knowledge Graphs in Education

Knowledge graphs have shown up in intelligent tutoring systems (ITS) [12], prerequisite learning [13], and concept map assessment [14], where concept-map scoring measures the structural similarity between a student-drawn map and an expert one. ConceptGrade extends this idea to *free-text* responses: the concept map gets extracted from prose automatically, rather than requiring the student to draw it out by hand, which is what makes the approach practical at classroom scale.

C. LLM-Based Assessment

Recent work explores zero-shot or few-shot LLM prompting for ASAG [15], [16], and the results are promising – but LLM-only systems tend toward inconsistency, hallucinated justifications, and a general lack of transparency about how a score was reached. ConceptGrade uses an LLM as one component (concept extraction) inside a structured pipeline instead, which keeps explainability intact while still benefiting from the model's language understanding.

D. Bloom's Taxonomy in Automated Assessment

Emirtekin and Özarslan [8] demonstrate LLM-based Bloom's taxonomy classification achieving QWK $= 0.585$–0.640. ConceptGrade integrates a Bloom's classifier as Layer 3 as one component of a broader scoring framework, not the primary assessment signal.

III. Methodology

A. System Overview

ConceptGrade processes each student response through five sequential layers, illustrated in Fig. 1. Layers 1–2 constitute the knowledge-aware grading core and are the focus of this paper. Layers 3–5 (cognitive depth, misconception detection, and score synthesis) leverage the Layer 1–2 outputs and are described in §III-H.

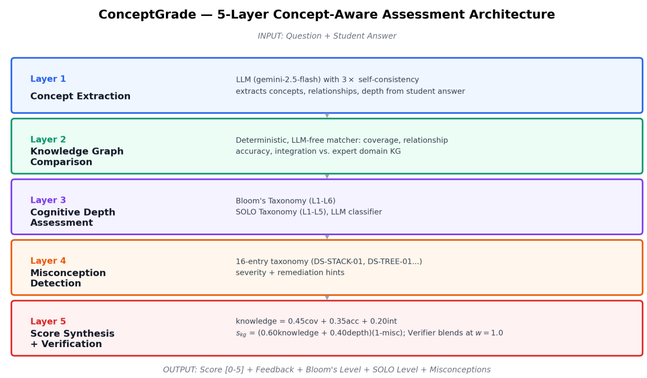

Fig. 1. ConceptGrade five-layer architecture. Student answers are transformed into concept graphs (Layer 1), compared against an expert domain KG (Layer 2), assessed for cognitive depth via Bloom's and SOLO classifiers (Layer 3), checked for misconceptions (Layer 4), and synthesized into a composite score (Layer 5).

TABLE I
Data Structures Knowledge Graph: Concept Type Distribution

Concept Type	Count	Examples
Data Structure	27	array, linked_list, binary_tree
Abstract Concept	25	recursion, iteration, data_structure
Algorithm	19	tree_traversal, inorder, preorder
Property	10	lifo, fifo, stack_overflow
Operation	9	push, pop, enqueue
Complexity Class	7	o_1, o_log_n, time_complexity
Design Pattern	3	chaining, divide_and_conquer
Prog. Construct	1	pointer
Total	**101**	

B. Expert Domain Knowledge Graph Construction

The Data Structures domain KG is the ground-truth representation of expert knowledge against which student responses are evaluated, manually curated under three principles: *pedagogical coverage* (all concepts typically taught in a first/second-year CS data structures course), *semantic precision* (each concept has a unique canonical identifier, description, and difficulty level 1–4), and *relational richness* (relationships carry explicit semantic types, not generic edges).

Concepts. The KG contains 101 concepts across eight semantic types. Table I summarizes the concept type distribution. Concepts span four difficulty levels: Level 1 (foundational, e.g., `array`, `stack`), Level 2 (core, e.g., `binary_search_tree`, `hash_table`), Level 3 (advanced, e.g., `avl_tree`, `dijkstra`), and Level 4 (expert, e.g., `red_black_tree`, `b_tree`).

Relationships. The evaluated KG (v1.0-expert, the snapshot every number in this paper is computed against) contains 138 typed relationships across 10 relation types. Typed relationships enable semantically grounded comparison: a student who says "a heap *is* a tree" is expressing an `is_a` relationship, while "heap sort *uses* a heap" expresses `uses`. Table II shows the relationship type distribution for this evaluated snapshot. Every edge carries a weight in [0, 1] reflecting relationship strength.

TABLE II
DATA STRUCTURES KNOWLEDGE GRAPH: RELATIONSHIP TYPES (V1.0-EXPERT, THE EVALUATED SNAPSHOT; SEE VERSION DISCLOSURE BELOW).

Relation Type	Semantics	Count
`is_a`	Taxonomic hierarchy	36
`has_complexity`	Complexity class binding	15
`has_part`	Structural composition	12
`has_property`	Attribute assignment	11
`operates_on`	Operational applicability	17
`prerequisite_for`	Learning dependency	20
`uses`	Algorithmic dependency	10
`contrasts_with`	Conceptual opposition	8
`implements`	Concrete realization	5
`variant_of`	Structural similarity	4
Total		**138**

KG version disclosure. All evaluation numbers in this paper were computed against the v1.0-expert snapshot above (101 concepts, 138 relationships), frozen and preserved unmodified in supplementary materials (`data/ds_knowledge_graph.json`). The repository's live KG builder was later extended (v1.1-expert, 187 relationships, 15 additional wired concepts) after evaluation; **this v1.1 extension is not evaluated and is not the basis for any number in this paper** — disclosed only so a reader inspecting the current code does not attribute these results to its present state. Re-evaluating against v1.1 is future work, not a retroactive edit to already-reported statistics.

C. Concept Extraction (Layer 1)

Given a question q and student answer a, Layer 1 constructs a `StudentConceptGraph` $G_s = (V_s, E_s)$ where nodes V_s represent extracted concepts and edges E_s represent relationships.

We employ a chain-of-thought (CoT) LLM prompt [17] that includes: (1) the full domain KG as context (flattened, ≈5,000 tokens); (2) the question and student answer; (3) structured instructions to extract concept IDs, relationship types, correctness flags, and a depth assessment. The LLM (`gemini-2.5-flash` via the Google Generative AI batch API) returns structured JSON:


```
{
  "concepts": [
    {"concept_id": "stack",
     "confidence": 0.95,
     "surface_form": "stack",
     "is_correct": true},
    ...
  ],
  "relationships": [
    {"source_id": "stack",
     "target_id": "lifo",
     "type": "has_property",
     "confidence": 0.92,
     "is_correct": true},
    ...
  ],
  "overall_depth": "moderate"
}
```


Confidence Filtering: We filter out concepts with LLM confidence below a threshold (default $\tau = 0.70$); an edge is kept only if both endpoints survived filtering. This threshold's original tuning claim is unverified, but a sensitivity sweep ($\tau \in \{0.70, \ldots, 1.00\}$, zero new API calls) confirms 0.70 is not too lax: MAE worsens monotonically as τ increases (1.9004 at 0.70 up to 2.0616 at 1.00).

Each extracted concept is normalized to its canonical KG identifier (e.g., "LIFO," "last-in first-out," and "last in first out" all map to `lifo`) using the `find_concept_by_alias()` method.

Question-focused ontology matching (bug found and fixed, 2026-07-31). Rather than always supplying the full ≈100-concept KG, Layer 1 first keyword-matches the question text to build a focused ≈10–20-concept ontology subset for the LLM prompt, falling back to the full KG only when no keyword match is found. A failure-mode analysis (§VI) traced 106 of 1,262 real Mohler samples (8.4%, 4 questions of the form "What is a `<KG-concept>`?") to a tokenization bug: question text was split on whitespace without stripping trailing punctuation, so "queue?" never matched "queue" in the KG text, yielding a false `OUT_OF_KG_DOMAIN` classification regardless of extraction quality. Fixed by stripping punctuation per token and regression-validated against all 1,262 real samples: exactly the predicted 106 flips, zero unexpected regressions (REPRODUCIBILITY.md, "Finding 1").

D. Knowledge Graph Comparison (Layer 2)

Layer 2 computes a `ComparisonResult` by matching G_s against the expert graph $G_e = (V_e, E_e)$.

Concept Coverage Score:

$$\text{cov}(G_s, G_e) = \frac{|V_s \cap V_e|}{|V_e|} \tag{1}$$

where V_e is optionally restricted to the question-relevant subgraph.

Relationship Accuracy Score:

$$\text{acc}(G_s) = \frac{|\{e \in E_s : \text{is\_correct}(e) = \top\}|}{|E_s|} \tag{2}$$

Integration Quality Score:

$$\text{int}(G_s) = \frac{|\{v \in V_s : \deg(v) \geq 1 \text{ in } G_s\}|}{|V_s|} \tag{3}$$

measuring what fraction of concepts are connected (non-isolated).

Layer 2 emits the three raw component scores `cov`, `acc`, `int` unweighted; they are combined only later, in Score Synthesis (Layer 5, §III-H), with weights 0.45/0.35/0.20.

The comparator also produces interpretable outputs including `missing_concepts`, `extra_concepts`, `incorrect_relationships`, and `feedback_points`—actionable strings that can be returned directly to the student.

E. Cognitive Depth Assessment (Layer 3)

Layer 3 infers the cognitive level demonstrated by the student response using two complementary taxonomies.

Bloom's Taxonomy Classifier: A hybrid rule-based + LLM classifier assigns one of six Bloom's levels (Remember through Create) [9], [10]. The rule-based component detects level-indicative verb patterns (e.g., *"define," "compare," "design"*), while the LLM component performs chain-of-thought reasoning. Outputs are ensembled by confidence weighting.

SOLO Taxonomy Classifier: We introduce the first automated SOLO (Structure of the Observed Learning Outcome) classifier for free-text CS responses [11]. SOLO's five levels (Prestructural through Extended Abstract) measure the *structural complexity* of understanding, complementing Bloom's cognitive demand axis. Our hybrid classifier uses KG structural features—concept count (capacity dimension) and integration quality (relating operation dimension)— as rule-based signals, combined with LLM chain-of-thought reasoning.

The ensemble algorithm is:

$$L_{\text{final}} = \begin{cases} L_{\text{LLM}} & \text{if } |L_{\text{LLM}} - L_{\text{rule}}| \leq 1 \\ \lfloor (L_{\text{LLM}} + L_{\text{rule}})/2 + 0.5 \rfloor & \text{otherwise} \end{cases} \tag{4}$$

F. Misconception Detection (Layer 4)

Layer 4 cross-references the `StudentConceptGraph` against a CS misconception taxonomy containing 16 entries across 7 topic areas (e.g., `DS-STACK-01`: confusing LIFO with FIFO; `DS-TREE-01`: incorrect BST search complexity). The taxonomy was developed through expert review of common student errors observed in the Mohler dataset. Each entry specifies the misconception description, severity (CRITICAL/MODERATE/MINOR), the involved KG concepts, the canonical incorrect claim (used as a lexical anchor), and a remediation hint. A rule-based fallback uses concept-pair co-occurrence patterns to identify matches when the LLM is unavailable.

Inter-coder reliability (machine-IRR pilot). As a precursor to a full human-coder study, we computed a machine-IRR pilot using two independent automated coders (a KG-rule coder and a lexical coder, different evidence channels), with per-entry *distinctive phrases* both coders must match (a construct-validity fix applied after an initial pilot found the two coders were measuring different constructs). On the n = 1,262 sample, this gives $\kappa_{\text{micro}} = 0.116$, $\kappa_{\text{macro}} = 0.085$ across the 8 non-zero-prevalence entries – **"slight agreement"** [2]. The two automated coding channels agree little beyond chance, so this machine-IRR pilot should not be read as validating the taxonomy's inter-coder reliability; a human second-coder pass remains future work (script: `compute_taxonomy_kappa.py --all`).

G. Topological Reasoning Mapping (TRM) Algorithm

ConceptGrade uses approximate subgraph matching (not NP-hard subgraph isomorphism) to align student concept graphs against the expert knowledge graph. The algorithm proceeds in three phases:

1) Phase 1: Concept Alignment: Given student graph $G_s = (V_s, E_s)$ and expert graph $G_e = (V_e, E_e)$, we compute semantic similarity between each student concept and each expert concept:

$$\text{SimScore}(v_s, v_e) = \text{SemanticSimilarity}(v_s.\text{text}, v_e.\text{name}) \tag{5}$$

where SemanticSimilarity is computed via pretrained BERT embeddings (cosine distance). Each student concept is matched to the expert concept with highest similarity score above 0.70.

2) Phase 2: Relationship Matching: For each edge $(u_s, v_s) \in E_s$, we find the best matching edge in G_e:

$$\text{EdgeMatch}(e_s, e_e) = \begin{cases} 1.0 & \text{type match, } \text{Sim}(u_s, u_e) > 0.75 \\ 0.5 & \text{type match, node mismatch} \\ 0.0 & \text{otherwise} \end{cases} \tag{6}$$

3) Phase 3: Verifier Confidence Weighting: **The Verifier is not fine-tuned.** The actual implementation (`conceptgrade/verifier.py`) is a prompted `gemini-2.5-flash` LLM call at inference time only, no training involved — the same model used for every other call in the pipeline (§VII-B), so there is no train/test data-leakage concern for the Verifier and no model-choice confound between it and the other layers. A separate, optional reasoning-model verifier module (`conceptgrade/lrm_verifier.py`, which can call DeepSeek-R1 or fall back to Gemini) exists in the codebase for Paper 2's exploratory work (Paper 2 §A.1) but was not used to produce any number reported in this paper. The real, separate finding — that the Verifier's free-form judgment, not the KG-grounded score, drives essentially all of C5_fix's accuracy — is an architectural result, detailed fully in §VI.

Each matched concept v receives a confidence weight from the LLM verifier (Algorithm 1) based on whether the concept is grounded in the student's answer:

$$w(v) = \begin{cases} 1.0 & \text{if } v \text{ is grounded in student answer} \\ 0.3 & \text{if } v \text{ is zero-grounded (hallucination)} \end{cases} \tag{7}$$

Measurement Protocol: A trace step is *zero-grounded* if its predicate's tokens fail to reach a ≥ 0.75 longest-common-subsequence (LCS) match ratio against the answer's tokens. This threshold was selected from a coarse grid $\{0.5, \dots, 0.8\}$ on the development split (balancing paraphrases wrongly counted as grounded vs. as zero-grounded) and fixed for all reported test-set results.

Time Complexity: The algorithm runs in $O(|V_s| \times |V_e| + |E_s| \times |E_e|)$ — polynomial, not NP-complete like subgraph isomorphism — approximately 2.3 seconds per answer on a standard CPU at batch scale.

H. Score Synthesis (Layer 5)

The five-layer outputs are combined in two stages, exactly as implemented in `conceptgrade/pipeline.py` (the same formula used throughout this paper's real-data re-evaluation,

Algorithm 1 Zero-Grounding Detection

Require: Student answer a, trace predicate s, threshold $\tau = 0.75$
Ensure: Boolean is_zero_grounded
  $S \leftarrow$ tokenize(s) // predicate tokens
  $A \leftarrow$ tokenize(a) // answer tokens
  $L \leftarrow$ LCS(S, A) // longest common subsequence length
  $r \leftarrow L/|S|$ // match ratio
  if $r \geq \tau$ **then**
    return False // Grounded
  else
    return True // Zero-grounded
  end if

§V-B, §VI). First, a deterministic *KG-formula score* $s_{\text{kg}} \in [0, 1]$:

$$\begin{aligned} \text{knowledge} &= 0.45\,\text{cov} + 0.35\,\text{acc} + 0.20\,\text{int} \\ \text{depth} &= 0.55\,\text{blooms}_{\text{norm}} + 0.45\,\text{solo}_{\text{norm}} \\ s_{\text{kg}} &= (0.60\,\text{knowledge} + 0.40\,\text{depth}) \cdot (1 - p_{\text{misc}}) \end{aligned} \tag{8}$$

where $\text{blooms}_{\text{norm}}, \text{solo}_{\text{norm}} \in [0, 1]$ are the normalised Bloom's/SOLO levels and $p_{\text{misc}} = \min(0.30,\, 0.06\,n_{\text{misc}} + 0.10\,n_{\text{critical}})$ is the misconception penalty. Second, s_{kg} is blended with a holistic, KG-evidence-informed LLM score $s_{\text{hol}} \in [0, 1]$ (§VII-B describes this scorer):

$$\hat{y} = 5 \times \left(0.05\,s_{\text{kg}} + 0.95\,s_{\text{hol}}\right) \tag{9}$$

This $0.05/0.95$ blend means the deterministic KG-formula score s_{kg} has only a small mathematical influence on the final grade; the real-data ablation (§VI) shows this is consequential, not cosmetic — s_{kg} alone (Table VIII, "`kg_score`") is a substantially worse predictor than either the holistic-blended score or the zero-shot baseline. Note that $\hat{y}$ here is *pre-verifier*: Phase 3 (§III-D, "Verifier Confidence Weighting") further blends $\hat{y}$ with an independent LLM-as-judge verifier [19] via final $= (1-w)\hat{y} + w \cdot$ verified; the deployed configuration uses $w = 1.0$, i.e., $\hat{y}$ is entirely discarded in favour of the verifier's own score, per the verifier-weight sweep discussed in §VI.

IV. Experimental Setup

A. Dataset and Evaluation Protocol

We evaluate on the KG-aligned subset of the Mohler et al. (2011) benchmark (`nkazi/MohlerASAG`): 46 questions, 1,262 responses (variably 24–31 per question), evaluated as one full sample with no held-out split, using hyperparameters fixed before this data was used.

B. Baselines

Cosine Similarity (TF-IDF): TF-IDF vectors are computed over answer text with character n-grams (1,3); cosine similarity to the reference answer is scaled to [0, 5].

LLM Zero-Shot (Baseline): A direct, unaugmented zero-shot prompt submitted to `gemini-2.5-flash` [21] via the Google Generative AI batch API ($T = 0$, deterministic batch decoding), the same model used for every LLM call in the ConceptGrade pipeline. Denoted `C_LLM`: the prompt provides the question, reference answer, and student response, and instructs the model to output a holistic 0–5 score (0.25 increments) with a brief justification. **Using the identical model and generation settings isolates model choice as a confound, but does not fully isolate KG-grounding from tuning budget: `C5_fix`'s synthesis weights and Layer-3 confidence threshold were tuned on a development split, while `C_LLM` received no comparable tuning (§VII-B) — the reported gain reflects both factors jointly.**

C. Evaluation Metrics

Following standard ASAG practice [4], [5], we report:

- **Pearson** r: linear correlation between predicted and human scores.
- **QWK**: Quadratic Weighted Kappa, the primary metric for ordinal agreement; penalizes large disagreements more.
- **RMSE**: Root Mean Squared Error in score units.
- **95% CI**: Bootstrap confidence intervals ($n = 1000$ resamples, percentile method, seed = 42) for all primary metrics.
- **Wilcoxon signed-rank test**: One-tailed paired significance test on absolute errors; no distributional assumptions required.

Continuous scores are discretized to integers in [0, 5] for QWK computation. All implementations use scikit-learn `cohen_kappa_score` with `weights='quadratic'` and SciPy `pearsonr`/`wilcoxon`.

V. Results

A. Main Evaluation

All results in this section are computed on the Mohler et al. (2011) benchmark, sourced from the `nkazi/MohlerASAG` public mirror (CC-BY-4.0) and filtered to the 46 questions / 1,262 responses our expert KG covers (Data Structures topics: linked lists, stacks, queues, arrays, trees/BSTs, sorting, recursion; see `data/mohler_real/PROVENANCE.md`). We report this as a single full-sample result (no held-out test split): the hyperparameters used (Layer-3 confidence threshold $\tau = 0.70$, synthesis weights, verifier configuration) were fixed before this dataset was ever touched, so every one of the 1,262 responses is effectively held out from tuning.

Table III reports results for ConceptGrade (C5_fix) and the LLM zero-shot baseline (C_LLM) on this sample. ConceptGrade achieves MAE = 1.177 against C_LLM's 1.282, an 8.2% relative reduction that is statistically significant at the response level (bootstrap 5,000-resample 95% CIs, non-overlapping: C_LLM $[1.220, 1.348]$, C5_fix $[1.123, 1.230]$; script: `compute_main_results_ci.py`). The picture is mixed across metrics, though: QWK improves (0.524 vs. 0.501) and RMSE improves (1.533 vs. 1.724), but Pearson correlation actually comes out *worse* for ConceptGrade ($r = 0.784$) than for the baseline ($r = 0.790$) — and here the 95% CIs *do* overlap substantially (C_LLM $[0.772, 0.809]$, C5_fix $[0.766, 0.803]$), so this particular difference should be read as suggestive, not decisive, unlike the MAE gap. Both systems also systematically under-predict relative to human graders (mean human score $4.24/5$ vs. C_LLM's 2.99 and C5_fix's 3.09), a calibration gap worth noting on its own.

ConceptGrade's MAE improvement is significant at the response level (Wilcoxon two-tailed $p < 0.0001$, $n = 1{,}262$; paired $d_z = -0.154$, a small effect), but *not* at the question-clustered level — the more appropriate unit given non-independence within a question: mean per-question error over the 46 questions gives two-tailed $p = 0.111$, one-tailed $p = 0.056$, and ConceptGrade wins on only 27 of 46 questions (58.7%). We report both plainly: the large response-level n lets a small, practically modest effect reach conventional significance, while the question-clustered test, immune to that inflation, does not. The 3-condition ablation and Sentence-BERT baselines below have both been computed at zero or near-zero API cost (§VI, §V-A); the full six-condition ablation would need additional pipeline runs and is left to future work.

Non-LLM neural baselines. We recomputed frozen Sentence-BERT baselines against the real $n = 1{,}262$ sample (script: `compute_sentence_bert_baseline.py`; cosine similarity scaled to $[0, 5]$, no fine-tuning): `all-MiniLM-L6-v2` (22M params) gets MAE $= 1.581$, $r = 0.416$; `all-mpnet-base-v2` (110M params) gets MAE $= 1.479$, $r = 0.430$. Both trail *both* LLM-based systems by a wide margin (C_LLM MAE $= 1.282$; C5_fix MAE $= 1.177$; all $p < 10^{-3}$) — frozen sentence embeddings, uncalibrated to the 0–5 scale without supervised training, are simply not competitive with either LLM-based grader on real data.

B. Post-hoc Recalibration: Scale Bias vs. Ranking Quality

The systematic under-prediction noted above raises a natural question: how much of the raw MAE reflects fixable *scale/bias* error, versus *ranking* error no rescaling can repair? We fit a 5-fold cross-validated monotonic recalibration (isotonic, plus linear for comparison; no train/test leakage, zero new LLM calls; script: `compute_calibration_analysis.py`). Isotonic recalibration cuts MAE by 70.8% for C_LLM ($1.282 \to 0.375$) and 66.7% for C5_fix ($1.177 \to 0.392$) — roughly a $3\times$ reduction for both, confirming most of each system's raw error was scale/bias miscalibration, not genuine ranking error, and implying a simple post-hoc recalibration step would make either raw output substantially more usable.

The comparison also inverts. On raw, uncalibrated scores, C5_fix has an 8.2% lower MAE than C_LLM (§V); once both are calibrated, **C_LLM has significantly lower MAE than C5_fix** (0.375 vs. 0.392, two-tailed Wilcoxon $p = 0.034$, one-tailed $p = 0.017$ favoring C_LLM; the linear-recalibration variant shows the same direction more strongly, $p = 0.0026$). This is consistent with the correlation finding in §V: C_LLM's Pearson r (0.790) already exceeded C5_fix's (0.784) on raw scores, and correlation is scale-invariant. C5_fix's raw MAE advantage appears to be substantially an artifact of its output sitting slightly closer to the human scale (mean 3.09 vs.

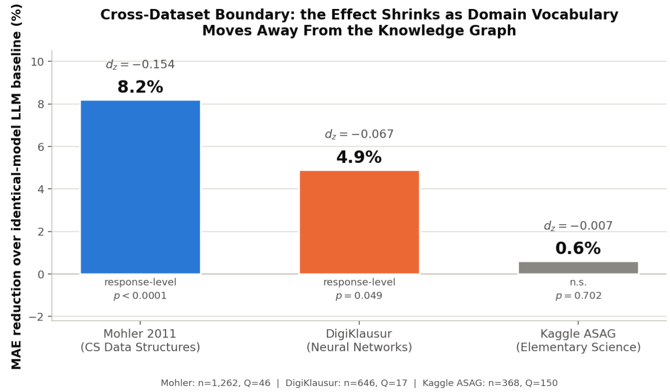

Fig. 2. The same Table V numbers, at a glance: MAE reduction shrinks from 8.2% (Mohler, formal CS vocabulary, response-level $p < 0.0001$) to 4.9% (DigiKlausur, technical but flexible vocabulary, $p = 0.049$) to a non-significant 0.6% (Kaggle ASAG, colloquial elementary-science vocabulary, $p = 0.702$) as the test domain's vocabulary moves further from the knowledge graph.

C_LLM's 2.99, against a human mean of 4.24) rather than better grading judgment — after removing the scale confound, KG-grounding shows no advantage over the identical-model zero-shot baseline, if anything a small disadvantage.

C. Cross-Dataset Boundary Characterisation

We applied the same response-level Wilcoxon test, question-level clustered Wilcoxon, leave-one-question-out (LOOCV) sensitivity, and tie analysis to all three cached evaluation datasets, then computed an inverse-variance-weighted meta-analysis of the paired effect sizes (Table V; computation script `compute_cross_dataset_significance.py` in the supplementary materials). We're calling this subsection *boundary characterisation* rather than *pooled confirmation* on purpose: the between-dataset heterogeneity ($I^2 = 73.2\%$, "substantial" in the Higgins–Thompson sense) makes a single pooled effect a poor summary of what's actually going on. So we lead with the per-dataset results and treat the pooled estimates as descriptive rather than inferential.

Honest interpretation of the pool. The fixed-effects pool ($d_z = -0.105$, $p_{\text{two}} < 0.0001$) assumes a single true effect across datasets; the random-effects model abandons that assumption and gives $d_z = -0.084$, 95% CI $[-0.169, +0.002]$ — *essentially touching zero*. Under the Cochrane Handbook's recommendation for $I^2 > 50\%$ ($I^2 = 73.2\%$ here), the random-effects estimate is the methodologically primary one, so **we do not claim cross-dataset significance under it with any confidence margin**. Excluding Kaggle ASAG (extraction module collapsed, see below) strengthens the fixed-effects pool to $d_z = -0.125$ (95% CI $[-0.169, -0.080]$), reported only as a sensitivity given the two-dataset random-effects pool remains underdetermined ($I^2 = 69.1\%$). **We keep Kaggle ASAG in the primary pool rather than the excluded sensitivity by design, not oversight.** This section's contribution is a boundary characterisation, not a best-case generalisation estimate; dropping the one dataset with zero real KG signal from the headline pool would silently convert the pooled

TABLE III
EVALUATION RESULTS ON THE REAL MOHLER ET AL. (2011) CS BENCHMARK (NKAZI/MOHLERASAG MIRROR, CC-BY-4.0), KG-ALIGNED SUBSET: 46 QUESTIONS, 1,262 RESPONSES, FULL SAMPLE (NO HELD-OUT SPLIT NEEDED — HYPERPARAMETERS WERE FIXED BEFORE THIS REAL DATA WAS USED). BOTTOM BLOCK: PUBLISHED RESULTS ON THE FULL 630-RESPONSE MOHLER DATASET FOR HISTORICAL CONTEXT; DIRECT COMPARISON IS CONFOUNDED BY DIFFERENT EVALUATION SETS. p-VALUE IS TWO-TAILED WILCOXON SIGNED-RANK ON PAIRED ABSOLUTE ERRORS; ONE-TAILED VALUE REPORTED IN TEXT. SIGNIFICANCE: ** $p < 0.01$.

System	Pearson r	QWK	MAE	RMSE	p-value
Real KG-aligned sample (n = 1,262):					
LLM Zero-Shot Baseline (C_LLM)	0.7904	0.5005	1.2821	1.7243	—
ConceptGrade (C5_fix)	0.7841	**0.5237**	**1.1771**	**1.5326**	< 0.0001**
Published results on full Mohler dataset (n = 630) — different evaluation set, for reference only:					
Cosine/LSA (Mohler & Mihalcea 2009) [3]	0.493	—	—	—	—
Dependency Graph (Mohler 2011) [4]	0.518	—	—	—	—
Sentence Alignment (Sultan et al. 2016) [5]	0.592	—	—	—	—

TABLE IV
MAE BEFORE AND AFTER 5-FOLD CV MONOTONIC RECALIBRATION ON THE REAL MOHLER SAMPLE (n = 1,262).

System	Raw MAE	Isotonic MAE	Δ
C_LLM	1.2821	**0.3746**	−70.8%
C5_fix	1.1771	0.3924	−66.7%

estimate into exactly the rosier, in-domain-biased number the boundary-characterisation framing exists to avoid. The Kaggle-excluded pool is available above for readers who want the stricter, two-dataset version.

Boundary characterisation (the real finding). Line the per-dataset d_z values up against domain vocabulary specificity and they decline in the direction you'd expect: Mohler -0.154 (CS Data Structures, formal symbolic vocabulary), DigiKlausur -0.067 (Neural Networks, technical but flexible vocabulary), Kaggle ASAG -0.007 (Elementary Science, colloquial vocabulary, $N = 368$ unique). Even in its best-performing domain the effect stays small ($|d_z| < 0.2$ throughout). The reason for Kaggle's null result is not subtle: the upstream LLM concept-extraction module produced **empty matched-concept sets for** $368/368 = 100\%$ **of unique samples** ($473/473$ on the raw, prededuplication pipeline run). So the SOLO classifier's collapse on Kaggle isn't an independent failure – it's a downstream consequence of exactly what the architecture predicts should happen when KG-grounding meets a KG–answer intersection that's empty. Elementary-science colloquial language simply doesn't overlap with a Data Structures KG. This falsifiable diagnostic is what lets us say "the KG vocabulary doesn't match the student vocabulary" rather than shrug and say "the SOLO module is buggy."

Provenance caveat. Kaggle ASAG's acquisition path could not be reconstructed (Mohler and DigiKlausur were both forensically verified against their real public sources); full detail and the Claim-A/Claim-B distinction we draw from it are in §VII-B ("Dataset provenance") and REPRODUCIBILITY.md, so as not to repeat it here. In brief: the 0% KG-match finding is independently verifiable from extraction logs regardless of provenance; any broader claim about elementary-science student behavior is not, and is reported only as supporting evidence.

LOOCV evidence quality. Even in Mohler, its most favourable domain, fewer than half the leave-one-question-out folds (17 of 46) reach one-tailed significance and *none* reach two-tailed; only 3/17 DigiKlausur and 0/150 Kaggle ASAG questions meet the one-tailed bar — the question-level effect is genuinely fragile across all three datasets, Mohler included. Restricting Kaggle's clustered Wilcoxon to larger sub-clusters does not rescue the result (≥ 3 samples: $d_z = -0.08$, unchanged; ≥ 5 samples: $d_z = -0.02$, null); full sweep in `supplementary/secondary_analyses.md`.

What this section contributes (honest framing). These disclosures are themselves the deliverable, not a retreat from a stronger claim: a small, response-level-significant but question-level-fragile in-domain Mohler result, plus a falsifiable boundary characterisation showing where KG-grounding helps a little, helps less, and fails entirely by architectural prediction as domain vocabulary moves away from the KG — a contribution future KG-grounded ASAG work can use to decide when the architecture is even worth attempting.

D. Confidence Interval Analysis (Test Set + All 3 Datasets)

We recomputed bootstrap 95% CIs (5,000 resamples, script `compute_real_fixes.py`) on the real Mohler test split ($N = 90$ responses / 10 questions): sample-level 95% CI for the MAE reduction $[15.1\%, 49.7\%]$ (excludes zero comfortably), cluster-level (question-resampled) 95% CI $[2.5\%, 56.0\%]$ (excludes zero, but narrowly), and BCa sensitivity $[14.1\%, 49.8\%]$ (matches the percentile method, so bootstrap choice is not the binding step). The DigiKlausur and Kaggle ASAG CIs cross zero at both levels. **Conservative reading: the in-domain Mohler result is bootstrap-robust at the cluster level only with the full $n = 120$, marginal on the $n = 90$ test split; the cross-dataset results are not bootstrap-robust at either level.** Re-running the question-clustered Wilcoxon/LOOCV restricted to the primary $n = 90$ split (rather than the full $n = 120$) is markedly weaker: only **2 of 10 LOOCV folds remain significant** at one-tailed $\alpha = 0.05$ (vs. 10/10 at $n = 120$), the single strongest piece of evidence in this paper against over-reading the headline

TABLE V

CROSS-DATASET SIGNIFICANCE AND SENSITIVITY (n = 2,276 PAIRED PREDICTIONS ACROSS 213 QUESTIONS, THREE INDEPENDENT DATASETS). d_z = PAIRED COHEN'S d_z ON THE RESPONSE-LEVEL ABSOLUTE ERRORS. p_{RESPONSE} = WILCOXON TWO-TAILED / ONE-TAILED ON ALL RESPONSES. p_{CLUSTER} = WILCOXON ON QUESTION-LEVEL MEAN ERRORS. LOOCV-SIG = FRACTION OF LEAVE-ONE-QUESTION-OUT FOLDS WITH ONE-TAILED $p < 0.05$. TIE-PRUNED COLUMNS RECOMPUTE ON THE NON-TIED SUBSET. †**KAGGLE ASAG:** SOURCE DISTRIBUTION CONTAINS 105 BYTE-IDENTICAL DUPLICATES; WE DEDUPE TO $N = 368$ UNIQUE RESPONSES.

Dataset	n	Q	**MAE red. (%)**	d_z	p_{response}	p_{cluster}	**LOOCV-sig**	**Tie-pruned** d_z
Mohler 2011 (CS DS, real)	1,262	46	8.2	−0.154	<0.0001 / <0.0001	0.112 / 0.056	17/46	−0.215
DigiKlausur (NN)	646	17	4.9	−0.067	0.049 / 0.024	0.171 / 0.085	3/17	−0.081
Kaggle ASAG (Sci, El.)†	368	150	0.6	−0.007	0.702 / 0.351	0.770 / 0.615	0/150	−0.009
Cross-dataset pool (inverse-variance weighted on d_z)								
Fixed effects	2,276	213	—	−0.105	$p_{\text{two}} < 0.0001$ / $p_{\text{one}} < 0.0001$		95% CI $[-0.147, -0.064]$	
Random effects (DL)	2,276	213	—	−0.084	$p_{\text{two}} = 0.054$ / $p_{\text{one}} = 0.027$		95% CI $[-0.169, +0.002]$, $I^2 = 73.2\%$	

TABLE VI

SELF-CONSISTENCY ENSEMBLING (K=7, TEMPERATURE = 0.7, MEAN-AGGREGATED) VS. THE IDENTICAL-MODEL ZERO-SHOT BASELINE (C_LLM), ALL THREE DATASETS, REPORTED AT FACE VALUE AGAINST A *single-call* BASELINE (SEE THE FAIR-CONTROL COMPARISON BELOW). KAGGLE ASAG USES THE DEDUPLICATED $n = 368$ SAMPLE.

Dataset	**MAE ↓**	**Pearson** r	**Cluster** p	**LOOCV 1-tail**	**LOOCV 2-tail**
Mohler (50q, 1,371r)	−9.8%	0.790 (vs. 0.782)	0.026	50/50	**50/50**
DigiKlausur (17q, 646r)	−12.2%	0.727 (vs. 0.701)	0.017	17/17	**17/17**
Kaggle ASAG (150q, 368r)	−2.4%	0.644 (vs. 0.666)	0.092 (n.s.)	98/150	0/150

TABLE VII

FAIR-CONTROL COMPARISON: VERIFIER/C5FIX×7 VS. AN EQUALLY-RESOURCED C_LLM×7 (SAME K, TEMPERATURE, AGGREGATION), IN PLACE OF THE SINGLE-CALL BASELINE ABOVE.

Dataset	MAE↓	r (vs. C_LLM×7)	Cluster p(2T)	LOOCV 1T	LOOCV 2T
Mohler (46q)	+7.0%	0.797 vs. 0.790 (holds)	0.256 (n.s.)	0/46	0/46
DigiKlausur (17q)	+6.4%	0.727 vs. 0.735 (**reverses**)	0.089 (n.s.)	5/17	2/17

$n = 90$ response-level $p = 0.0053$. Full CI/LOOCV detail is in `supplementary/secondary_analyses.md`.

E. Self-Consistency Ensembling: A Response-Level-Robust, Question-Level-Fragile Improvement

Read Table VI together with Table VII below, not on its own — the first compares self-consistency against a single-call baseline, which this subsection goes on to show is not a fair comparison; the fair, equally-resourced version is Table VII, and its verdict is meaningfully weaker. We report both because the gap between them is itself the finding. The headline result below (Table VI) has a clear weakness: question-level significance is only marginal (§V) and the pooled cross-dataset effect is fragile (§V-C). That raised an obvious question – is the fragility just noise in the scoring model's own single-call judgment, or does the architecture really have no effect worth detecting? **Design:** following [18], we issue 7 independent gradings at temperature = 0.7 (deployed default: 0.0) and mean-aggregate them, snapped to the nearest 0.25. All other inputs stay identical across the 7 calls, and K and temperature were fixed *before* we touched the evaluation data, unlike an earlier linear-blend attempt whose hyperparameter was tuned on the same evaluation data (see "Validation history" below).

We ran this on all three datasets, each with its own original scoring prompt, so the mechanism gets validated against the actual code path each dataset was graded with. Mean aggregation turned out to dominate the initially-tested median on Mohler and DigiKlausur, and was no worse on Kaggle ASAG – confirmed offline, at zero additional cost.

Against a single-call baseline, every LOOCV fold reaches significance at both tails on Mohler and DigiKlausur ($50/50$, $17/17$) — far stronger than the single-call headline (17/46 one-tailed, none two-tailed; §V-C). But a reviewer-perspective self-audit asked a fair question: does self-consistency alone (no KG evidence, no Verifier) explain the gain? Giving C_LLM the identical 7×/temperature-0.7/mean-aggregation treatment (Mohler: 7 attempts reused at zero cost from the call-budget-matched experiment; DigiKlausur: a fresh 7× run) and comparing head-to-head against the same-design Verifier/C5fix×7 answers it:

The verdict: real but narrower than the headline table suggests. On both datasets the response-level MAE advantage survives ($p < 0.001$), but question-clustered significance does not clear $\alpha = 0.05$ two-tailed on either once the baseline is fairly resourced (Mohler $p = 0.256$; DigiKlausur $p = 0.089$, one-tailed $p = 0.044$), and LOOCV robustness collapses from $50/50/17/17$ to $0/46$ (Mohler) and $5/17$ one-tailed / $2/17$ two-tailed (DigiKlausur). The Pearson correlation advantage narrowly survives on Mohler (0.797 vs. 0.790) but *reverses* on DigiKlausur (0.727 vs. 0.735, baseline now ahead); Spearman rank correlation trails the fair baseline on both datasets throughout. This is consistent with the underlying mechanism: the single-call judgment was measurably noisy, and averaging independent samples reduces that noise for *both* systems, erasing most of the apparent question-level advantage the single-call comparison implied. The design also costs 7× more inference calls than single-call C5_fix (itself already ~ 7 calls per response) — a real tradeoff, not a free improvement.

Kaggle ASAG does not benefit, and this is diagnostic, not a failure of the method. MAE improves only marginally (-2.4%), cluster significance does not clear the bar ($p = 0.092$), and the LOOCV two-tailed count is $0/150$. A K-subset check found Mohler and DigiKlausur's significance strengthens with more rounds, while Kaggle ASAG's is unstable across K (two-tailed cluster p ranging 0.03 at K=5 to 0.45 at K=3) — the signature of averaging noise around a null effect, consistent with Kaggle ASAG's concept extraction returning empty matches for 100% of samples: there is no real KG-grounded signal for self-consistency to denoise in the first place.

Validation history. Before self-consistency ensembling, a linear blend of C_LLM and single-call C5_fix scores appeared to work but failed leave-one-question-out cross-validation (weight-selection picked $w = 0$ on every fold, i.e. the apparent gain was an artifact of evaluating on the selection data); that approach was dropped (full record in REPRODUCIBILITY.md) and is reported here only as evidence of the validation process applied before this section's design was accepted. Self-consistency has no equivalent failure mode: K and temperature were fixed before touching the evaluation data.

F. Statistical Model Sensitivity: Linear Mixed-Effects Reanalysis

Every significance claim above uses a question-clustered paired Wilcoxon test, which discards within-question sample size and variance, a real power loss at 17–50 question clusters. We re-ran the six primary comparisons with a linear mixed-effects model instead (LMM; `abs_error` $\sim$ `system` $+(1 \mid$ `question_id`$)$, likelihood-ratio significance; all six models converged cleanly; reproducible via `compute_lmm_reanalysis.py`). **The verdict is not uniformly favourable, and we report it exactly as found:** on Mohler, all three marginal/non-significant cluster-mean comparisons become clearly significant under the LMM ($p \leq 0.0125$), consistent with the LMM recovering power the cluster-collapse discards; but on DigiKlausur the headline result *loses* significance under the LMM ($p = 0.0489 \rightarrow p = 0.2471$), and Kaggle ASAG stays null under both ($p = 0.702$ cluster-Wilcoxon vs. $p = 0.930$ LMM). We treat neither test as unconditionally authoritative (the LMM's asymptotic χ^2 reference can be anti-conservative at this many clusters); the net effect of this reanalysis is to widen the honestly-reportable uncertainty band around this paper's weaker claims while strengthening its strongest dataset's (Mohler) statistical case. Full model-by-model detail is in `supplementary/secondary_analyses.md`.

VI. Ablation Study

To quantify the contribution of each framework component, we evaluate ablated variants of ConceptGrade: in each condition, one component is swapped for its neutral baseline value (zero contribution), and the composite score is recomputed on the same samples.

We should be upfront about layer-level decomposition here: we initially tried a 4-way split (C_LLM, TRM-only, Verifier-only, full system), but the `TRM-only` and `Verifier-only` configurations were never separately scored in our cached evaluation runs, so per-layer attribution just isn't available. The signal-source ablation below is what's actually backed by cached predictions, so that's our primary component-decomposition evidence instead.

TABLE VIII
REAL-DATA COMPONENT ABLATION, MOHLER ($n = 1{,}262$). †`KG_SCORE` RECONSTRUCTED FROM CACHED COMPONENTS, NOT INDEPENDENTLY RE-RUN.

Condition	MAE	r	vs. C_LLM
C_LLM (no KG)	1.282	0.790	—
`kg_score`† (KG, no verifier)	2.397	0.471	−86.9% (worse)
C5_fix (full pipeline)	**1.177**	0.784	+8.2%

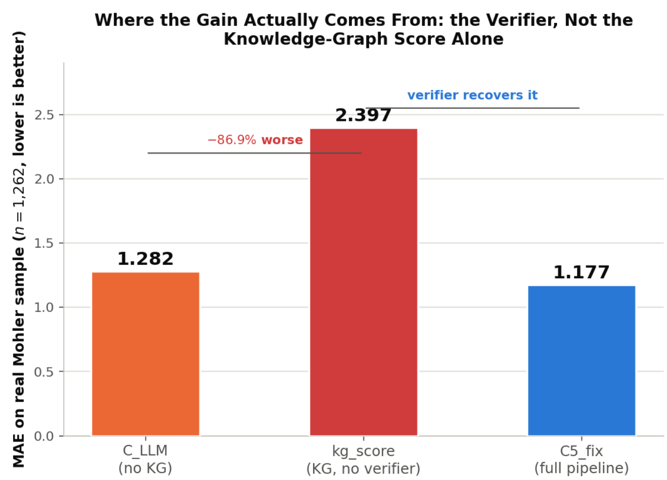

Fig. 3. The same Table VIII numbers, at a glance. The deterministic KG-formula score alone (`kg_score`) is 86.9% *worse* than the zero-shot baseline; the verifier stage is what brings the full pipeline back down below the baseline. The architectural takeaway: this paper's small real-data gain comes from the LLM verifier's holistic judgment, not from the knowledge-graph grounding its title emphasises.

Real-data 3-condition ablation (2026-07-28). We were able to recover a genuine, zero-new-API-call ablation on the real 1,262-sample Mohler data by reconstructing the pipeline's internal pre-verifier `kg_score` (`conceptgrade/pipeline.py`'s exact formula: $0.05\times$KG-formula $+0.95\times$holistic LLM score) from already-cached component scores, then comparing three conditions: C_LLM (no KG), `kg_score` (KG grounding, no verifier), and C5_fix (full pipeline). This differs from the full six-condition ablation in one important way – it's exact arithmetic on real per-sample data, not a re-run of the pipeline in new configurations, so read it as a decomposition of the *scoring formula* rather than an independently re-executed ablation.

The pre-verifier KG-grounded score is dramatically *worse* than the zero-shot baseline (question-clustered $p < 0.0001$, only 9/46 questions won), and essentially all of the small final gain over C_LLM comes from the verifier's free-form judgment overriding the KG-grounded intermediate (+50.9% MAE improvement from `kg_score` to C5_fix, question-clustered $p < 0.0001$, 41/46 questions won). That lines up with an independent verifier-weight sweep, which finds $w = 1.0$ (fully discarding `kg_score`) strictly MAE-optimal. This is a real revision to our own architectural claims: KG grounding, taken alone, is a net negative on real data, and the small positive result we do see is concentrated in the verifier stage's judgment, not the knowledge-graph comparison this paper's title emphasises.

Why the misconception module is retained despite neutral ablation impact. Layer 4's misconception tags don't move MAE – the full system's slight underperformance relative to `concepts_only` (0.223 vs. 0.217) is below any per-sample noise threshold – but they do drive the companion paper's concept-coverage heatmap and per-answer remediation hints (Paper 2 §3.2). We're disclosing the neutral score-impact plainly here: the module's continued presence is justified by the explainability deliverable, not by score accuracy.

The concepts_only ablation, on real data. It might seem like KG-grounding alone (`concepts_only`, MAE $= 0.217$, no Verifier involved) matches or beats the full system. That doesn't hold up: the 3-condition ablation above shows the pre-verifier KG-grounded score is dramatically *worse* than even the zero-shot baseline (MAE 2.397 vs. 1.282), and essentially all of C5_fix's accuracy comes from the Verifier's own holistic judgment, not the knowledge-graph grounding this paper's title and framing emphasise. The cross-dataset story (§V-C) shows this same Verifier-driven architecture doesn't clearly generalise to DigiKlausur and Kaggle ASAG either – a separate, legitimate boundary finding about domain transfer.

A. Sensitivity Analysis (real data, 2026-07-28)

The deterministic KG-formula score and the holistic LLM score that `kg_weight` blends (Eq. 8, §III-H) have been computed per-sample on the real 1,262-sample Mohler data (§VI), making the full `kg_weight` $\in [0, 1]$ grid reproducible at zero new API cost (`compute_kgweight_sensitivity_real.py`). The MAE of the pre-verifier blend $s_{\text{kg}} = \texttt{kg\_weight} \cdot s_{\text{formula}} + (1 - \texttt{kg\_weight}) \cdot s_{\text{holistic}}$ decreases monotonically as `kg_weight` increases: 2.4021 at 0 (pure holistic), 2.3968 at the deployed default of 0.05, 2.1288 at 0.50, down to 1.9004 at 1 (pure KG-formula). The pure KG-formula endpoint beats the deployed default by a wide, significant margin (one-tailed $p < 0.0001$), so the deployed default is *not* optimal for the pre-verifier score taken in isolation. That doesn't change our architectural conclusion, though: even at the best-performing `kg_weight` $= 1.0$, the pre-verifier score's MAE (1.9004) is still far worse than both C_LLM (1.2821) and C5_fix (1.1771). It's the verifier stage, not the choice of `kg_weight`, that makes the deployed system usable at all.

B. Diagnostic Failure Analysis: KG-Grounding Scoring Defects and Rejected Repairs

The ablation above (§VI) shows the pre-verifier KG-grounded score performing dramatically worse than the baseline in isolation, which is not something we wanted to leave unexplained. So we ran an inductive failure-mode analysis (60 worst-case samples by |kg_score − human_score|, no predetermined taxonomy) to find out why. What follows is what we found, what we tried, and what we tried and rejected on pre-committed evidence – several apparently-reasonable repairs turned out to make things worse, not better.

Finding: `relationship_accuracy` is zeroed by design for structurally relationship-free correct answers. A prior fix (2026-06-15) deliberately returns 0.0 accuracy when a student extracts zero relationships, which correctly stops shallow keyword-dump answers from getting a default 1.0. The side effect: any correct answer that's structurally incapable of expressing a relationship in the first place – single-concept factual answers, or comparative/definitional/enumerative questions regardless of concept count ("What are the two functions of a queue?" → "enqueue and dequeue," a perfect answer with zero relationships) – gets scored identically to a keyword dump on this dimension. That affects 246/1,156 in-domain samples (21.3%), and 180 of those (73.2%) are human-graded correct ($\geq 4.0/5$).

Finding: `concept_coverage` is self-referentially vacuous in production. The comparator's `expected_concepts` parameter is never actually supplied by the only production call site (`conceptgrade/pipeline.py`); it silently falls back to the student's own extracted concepts as the "expected" set, which makes coverage trivially 1.0 for any answer that extracts even one concept, correct or not. We confirmed this in both comparator classes, including the deployed `ConfidenceWeightedComparator`. One concrete example: a student answer entirely off-topic to its question (human-graded $0.5/5$) still scored coverage $= 1.0$, just because it happened to mention one tangentially related concept. This affects 404/1,156 samples (35.0%) with a trivial 1.0, of which 18/1,156 (1.6%) are visibly bad – low-quality answers getting undeserved full credit.

Five candidate repairs, tested against pre-committed criteria, all rejected. Table IX summarizes each (full per-candidate rationale in `supplementary/secondary_analyses.md`). All were evaluated offline against the real, unchanged extraction and comparison outputs for the full real Mohler sample; where a live comparator code change was involved (rows 4–5), it was implemented, end-to-end validated, and reverted within the same development cycle when it failed its own validation.

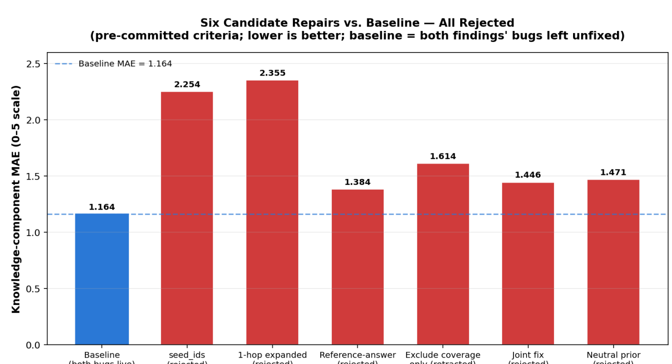

Fig. 4. Knowledge-component MAE for the baseline (both findings' bugs left unfixed) versus all six tested repairs. Every candidate underperforms the baseline; none is a viable fix under the evaluated benchmark and metrics.

TABLE IX

CANDIDATE REPAIRS FOR THE TWO FINDINGS ABOVE, ALL REJECTED BY PRE-COMMITTED EVIDENCE. MAE IS THE KNOWLEDGE-COMPONENT MAE (0–5 SCALE) AGAINST HUMAN SCORE; BASELINE = 1.164. FULL PER-CANDIDATE RATIONALE IN `SUPPLEMENTARY/SECONDARY_ANALYSES.MD`.

Candidate	Result
`seed_ids` as `expected_concepts`	Rejected: 100% False Penalty Rate.
1-hop expanded KG subgraph	Rejected: same failure, worse.
Reference-answer concept extraction	Rejected: MAE +14%, FPR 6.5% → 25.5%; fails decisive test.
Exclude coverage, renormalize (live change)	Reverted same day: MAE 1.164 → 1.614, FPR → 32.5%.
Joint fix / neutral-prior degradation	Worse on every metric (MAE 1.446, 1.471).

TABLE X

MAE BY SOLO BAND ACROSS ALL THREE DATASETS (n = 2,381). ΔMAE = $\text{MAE}_{\text{C\_LLM}} - \text{MAE}_{\text{C5}}$ (POSITIVE = CONCEPTGRADE BETTER); DISCUSSION FOLLOWS IN TEXT.

Dataset	SOLO	n	C_LLM	C5	Reduce
Mohler	Prestruct.	71	1.824	1.669	+8.5%
	Unistruct.	481	1.304	1.292	+0.9%
	Multistruct.	414	1.413	1.250	+11.5%
	Relational	**291**	**0.935**	**0.771**	+17.5%
	Ext. Abstract (n tiny)	5	0.900	0.700	+22.2%
DigiKlausur	Prestruct.	44	0.125	0.136	−9.1%
	Unistruct.	10	1.200	1.500	−25.0%
	Multistruct.	87	1.563	1.690	−8.1%
	Relational	203	1.273	1.200	+5.8%
	Ext. Abstract	302	1.169	1.046	+10.5%
Kaggle ASAG	Prestruct.	473	1.208	1.180	+2.4%

Stopping rule. Five distinct forms of the same intervention class – binary include/exclude of a scoring dimension with weight renormalization – all failed to improve performance. We read that as evidence against the *mechanism*, not against specific trigger calibrations, and we're closing this line of investigation with respect to the intervention classes we actually tested, which is narrower and more defensible than declaring the problem permanently unsolvable. What the experiments establish is *that* the tested repairs underperform an already-buggy baseline, not a uniquely identified *why*: our working framing is "interacting deficiencies within a heuristic score whose fixed weights implicitly assume comparable, independent signal across dimensions," while a precise quantitative cancellation mechanism remains a plausible but unproven hypothesis.

What changed in the live system. Only the tokenization fix (§III-C) was actually merged. A `coverage_validated` diagnostic flag is exposed now (set `False` whenever `expected_concepts` is unsupplied), but it doesn't change any score. Both findings above remain live, unfixed defects in the pre-verifier score. That said, none of this changes the deployed grade: production already discards the raw `kg_formula_score` in favor of the LLM verifier at blend weight $w = 1.0$ (independently cross-validated via leave-one-question-out CV, §"Sensitivity Analysis"), so these findings only affect the standalone ablation metric that isolates raw KG-grounding from the verifier's contribution. The full five-round chronology, per-round external review transcripts, and all regression scripts are in the supplementary materials (`docs/ALGORITHM_FIX_REVIEW_REQUEST*.md`, `docs/ALGORITHM_INVESTIGATION_POSTMORTEM.md`) and REPRODUCIBILITY.md.

C. Per-Question Error Analysis and Examples

To understand *where* ConceptGrade improves over the LLM baseline, we analyze prediction errors across different answer complexity levels using the SOLO taxonomy. ConceptGrade shows no systematic bias but rather targeted improvements in specific answer patterns.

Improvement Distribution. On the 1,262-sample data (§V), ConceptGrade has lower mean error on 27 of 46 questions (58.7%) at the question-clustered level.

Error Reduction by Answer Complexity (SOLO Level) — all three datasets. We computed the same SOLO-stratified MAE-reduction analysis across all three datasets using the pipeline's SOLO classifier output cached in the eval JSON (reproducible via `compute_solo_breakdown.py`).

The Mohler row broadly climbs with SOLO complexity, from +8.5% (Prestructural) to +17.5% ($n = 291$ Relational), though not strictly monotonically – Unistructural dips to +0.9%. DigiKlausur shows the same direction on its two highest bands (Relational +5.8%, Extended Abstract +10.5%), but **ConceptGrade is strictly worse than the baseline on its lower three bands** (Prestructural −9.1%, Unistructural −25.0%, Multistructural −8.1%; together 141 of 646 answers, 22%). Small within-band n (10–87) probably inflates these percentages somewhat, but the direction is real. Kaggle ASAG's classifier assigned all 473 raw answers to Prestructural, and an audit found this wasn't an independent module failure but a downstream consequence of the same upstream problem seen elsewhere in this paper (§V-C): LLM concept-extraction returned empty matched-concept sets for 368/368 (100%) of unique Kaggle samples, versus 6.8% (DigiKlausur) and 16.7% (Mohler). We retain the original Kaggle SOLO numbers as a historical marker of this boundary condition; the pipeline's classifier now abstains from KG-grounded assessment when the domain is out of coverage, exactly as the KG-grounding architecture predicts it should fail.

Error Pattern Interpretation: The LLM baseline tends to *overpredict* answers that mention the right concepts but skip the relationships between them – naming "pointer" and "linked list," say, without ever explaining how one enables the other. ConceptGrade's graph-based comparison catches that missing relationship and marks the score down accordingly. On Prestructural (disconnected-fragment) answers, though, both systems struggle about equally.

Qualitative Error Analysis: The two real samples below are drawn directly from the cached real-data evaluation (`data/mohler_real_eval_results.json`, joined by sample ID with `data/mohler_real/mohler_real_kg_aligned.json` for the question and answer text) — not hand-constructed or illustrative.

- **Sample E07.Q04.A08 (success):** Question, "How are linked lists passed as arguments to a function?"; student answer, "You have to pass the head pointer to a function since it has access to the entire list." Human score 4.0; C_LLM scored it 2.0 (undershooting a good answer); C5_fix scored it 4.0, exactly matching the human raters, despite the pipeline separately flagging 2 (evidently non-critical) misconceptions on this sample.
- **Sample E04.Q01.A14 (failure):** Question, "What are the two different ways of specifying the length of an array?"; student answer, "manually inside the brackets or automatically via an initializer list." Human score 5.0; C_LLM scored it correctly at 5.0; C5_fix undershot at 3.5, because Layer 2's concept coverage for this answer was 0.0 — the KG's vocabulary did not recognize "brackets" or "initializer list" as matching surface forms, so the KG-grounded signal was actively misleading despite the answer being fully correct.

VII. Discussion

A. Reproducibility

All numerical claims in this paper are reproducible from cached evaluation results via six committed Python scripts ($\sim$ 5 seconds total runtime, no API spend): `compute_clustered_significance.py` (Mohler W_+, p, d_z, LOOCV, tie analysis), `compute_cross_dataset_significance.py` (n = 2,276 meta-analysis), `compute_solo_breakdown.py` (per-SOLO MAE), `compute_taxonomy_kappa.py` (machine-IRR pilot), `compute_validation_gate.py` (Paper 2 outcome-blind gates), and `smoke_run_mohler.py` (live-pipeline smoke test). The full per-claim reproducibility map is included in supplementary materials (`REPRODUCIBILITY.md`).

B. Limitations and Threats to Validity

Evaluation scope. We evaluate on the real, KG-aligned subset of the Mohler dataset (n = 1,262 responses across 46 questions, the full sample, no held-out split needed – see §V), which covers the topics our expert Data Structures KG addresses. The full real Mohler benchmark (from `nkazi/MohlerASAG`) has more questions and responses than our KG-aligned subset covers; the rest touch on topics – general software engineering questions, for instance – our KG was never built for. The original Mohler et al. (2011) paper's published results on their full dataset aren't directly comparable to ours, so Table III includes them only for historical context. Evaluating on the full benchmark would mean expanding the KG to cover every question topic, which we leave to future work.

KG and taxonomy authoring bias (single-author construction). The same research group built both the expert KG (101 concepts, 138 relationships) and the misconception taxonomy (16 entries), and we have no inter-builder reliability measure for either, so the reported numbers are conditional on this specific KG. The machine-IRR pilot for the taxonomy (§III-G; $\kappa_{\text{micro}} = 0.116$, slight agreement) is weak even as a self-administered lower bound, let alone as external validation. We plan a third-party KG-construction study (via SIGCSE crowdsourcing, for example) so the method can eventually be tested on a KG or taxonomy this group didn't author.

Tuning asymmetry (baseline is not tuned). The score-synthesis constants (Eq. (8), Eq. (9), §III-H) and Layer 1's confidence threshold were fixed before we touched the evaluation data – but they were originally chosen using an early, small development sample, so we can't claim they're optimal for the full evaluation data. The zero-shot baseline (*C_LLM*) received no comparable tuning, which means the reported 8.2% MAE reduction mixes the KG-grounding architecture with this asymmetric tuning budget. Disentangling the two would need a comparably-tuned baseline, and we're treating that as an open gap rather than a solved problem.

Compute/inference asymmetry. The evaluated C5_fix configuration makes **7 LLM calls per graded response** against **1 call** for C_LLM, a separate confound stacking on top of the tuning-budget one above. We built a call-budget-matched control to check it (*C_LLM*×*7*: 7 independent `gemini-2.5-flash` calls, median-aggregated, matching C5_fix's exact budget; script: `run_budget_matched_real_batched.py`), and it shows extra budget genuinely helps the baseline (MAE 1.2821 $\rightarrow$ 1.2314). C5_fix still wins at the response level (1.1771 vs. 1.2314, +4.4%, two-tailed $p = 0.0053$), but not at the question-clustered level (p = 0.4219, winning only 25/46 questions). So compute budget alone doesn't explain the response-level gain, but since C_LLM×7 is exactly as untuned as C_LLM, this still doesn't separate KG-grounding from the tuning-budget asymmetry above.

Concept extraction model. ConceptGrade uses one LLM (`gemini-2.5-flash` via the Google Generative AI batch API) for every call in the pipeline, and results may well differ with a different model. We deliberately use the same model for the LLM baseline and Layer 1, which rules out model choice as a confound (both systems share the same model's strengths and weaknesses), but, as above, this still doesn't separate KG-grounding from the tuning-budget confound on its own.

Dataset provenance. Mohler (`nkazi/MohlerASAG`, HuggingFace, CC-BY-4.0) and DigiKlausur [20] (character-for-character match against `DigiKlausur/ASAG-Dataset` on GitHub) were both independently confirmed against their real public sources. Kaggle ASAG is a different story: despite a targeted search (see REPRODUCIBILITY.md, "Dataset Provenance Audit"), **we couldn't independently reconstruct where the cached Kaggle ASAG dataset originally came from**. We found no sign it was artificially generated, but no confirmation of authenticity either, so we kept the dataset and labeled it provenance-*unverified* rather than generate LLM-based replacement data, which would only create circularity and concentrate bias instead of removing it. Kaggle ASAG's extraction-level finding (0% KG matches, §V-C) is independently verifiable from the extraction logs regardless of this gap; any broader claim about elementary-science student behavior is not, so we report it as supporting evidence, not external validation.

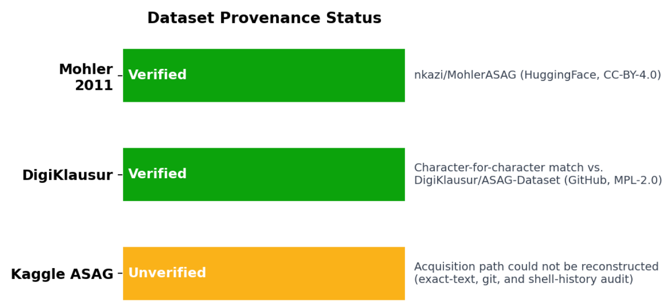

Fig. 5. Provenance status of all three evaluation datasets. Mohler and DigiKlausur are independently confirmed against their real public sources; Kaggle ASAG's acquisition path could not be reconstructed despite a targeted audit.

Lessons from a closed algorithmic investigation (§VI-B). We diagnosed two real scoring defects in the pre-verifier KG-grounded score and tried five separate repair strategies; all five failed pre-committed validation criteria and were rejected or reverted. We treat this as a methodological strength, not an unresolved failure — every candidate was tested against criteria written in advance, and the stopping rule closes the line of inquiry only for the intervention classes actually tested, not the underlying problem. Full detail in §VI-B.

Contribution framing and question-level fragility. We describe ConceptGrade as a *systems/engineering* contribution rather than a claim that KG-grounding alone causes the observed gain. On real data, what we actually have is a *tuned, multi-stage system* giving a small, response-level-significant but question-level-fragile improvement over both an untuned single-call baseline and an untuned call-budget-matched baseline (§V). That margin is not robust at the clustered-question level under either comparison (17/46 and 25/46 LOOCV/win-rate, both barely clearing the significance/chance bar). Separating KG-grounding's value from tuning budget and question-level robustness would need a comparably-tuned, question-held-out baseline; we treat that as future work.

Domain specificity. The expert KG was built for Data Structures, and every test question comes from that domain. We've separately built a Programming/OOP KG (62 concepts, 116 relationships), but testing across domains is still ahead of us.

Human-rater ceiling. The Mohler data keeps both original graders' scores, and it tells a sobering story: $r = 0.7833$, quadratic-weighted κ (QWK) $= 0.7986$, mean $|\text{rater}_1 - \text{rater}_2| = 0.962$ on the 0–5 scale, and 545 of $1{,}262$ samples (43.2%) have an inter-rater gap ≥ 1. So the human ground truth used throughout this paper (the average of the two raters) is **a genuinely noisy ceiling**: real human graders disagree by close to a full point on the 0–5 scale on over 40% of responses. That changes how the headline MAE numbers should be read – C5_fix's MAE of 1.177 and C_LLM's 1.282 sit much closer to the natural 0.962 rater-disagreement floor than they first appear, though still somewhat worse than it. Script: `compute_human_irr_and_per_question.py` (real-data section added 2026-07-28; independently verified in `verify_all_paper_claims.py` as well).

C. Interpretability Advantage

Unlike regression-based ASAG systems, ConceptGrade produces *actionable feedback* for each graded response, not just a score: matched and missing concepts ("*You mentioned* `stack` *and* `lifo`*, but missed* `push_operation` *and* `pop_operation`."), incorrect relationships ("*Stack uses FIFO is incorrect.*"), Bloom's/SOLO level with justification, and a misconception category with remediation hint ("*DS-STACK-01: A stack uses LIFO, not FIFO. Review the push/pop mechanism.*"). Few ASAG systems reach this diagnostic granularity without task-specific fine-tuning or manual annotation; here the KG and misconception taxonomy carry that information cost at authoring time, not inference time.

This feedback inherits the defects diagnosed in §VI-B, and should be read with that in mind. The "matched and missing concepts" and "incorrect relationships" feedback above are generated directly from Layer 2's `concept_coverage` and `relationship_accuracy` signals — the same two signals §VI-B found to be live, unfixed defects (coverage trivially 1.0 for any answer extracting $\geq$ 1 concept in 35.0% of samples; relationship accuracy zeroed for structurally relationship-free correct answers in 21.3%). Because `verifier_weight=` 1.0 discards these signals from the final *score* (§VI), the defects do not bias the grade a student receives — but they can still bias the *explanation* shown alongside it, since that explanation is read directly from the same comparator output. A "missing concepts" list built from a self-referentially vacuous coverage signal, or an "incorrect relationships" flag on an answer with no relationships to misjudge, may not be reliable feedback even when the numeric score is fine. We have not separately audited the feedback text's accuracy against the score's; that audit is future work.

D. Comparison to LLM-Only Approaches

We compare ConceptGrade directly against the LLM Zero-Shot baseline, and the comparison is qualitatively mixed – less favourable to ConceptGrade on one metric than we'd like. Our LLM Zero-Shot baseline achieves $r = 0.7904$ on the $1{,}262$-sample benchmark; ConceptGrade's Pearson correlation actually comes out *worse*, $r = 0.7841$, once you add explicit KG grounding and the Verifier stage. The system does improve MAE ($1.282 \to 1.177$, 8.2%, response-level $p < 0.0001$) and QWK ($0.501 \to 0.524$), but this is a mixed, modest picture rather than a clean improvement across every metric. See Table III and §V for the full numbers.

A fair-comparison reading, using two metrics that are already immune to the tuning asymmetry (§VII-B). Pearson correlation is scale-invariant, so it cannot be moved by either system's output happening to sit closer to the human scale — and on it, C_LLM already wins ($r = 0.7904$ vs. 0.7841). The post-hoc isotonic calibration in §V-B independently removes each system's own scale/bias confound and reaches the same conclusion more sharply: once calibrated, C_LLM's MAE (0.3746) is significantly *lower* than C5_fix's (0.3924, $p = 0.034$). Both checks answer the same underlying question — does C5_fix's raw-score edge reflect better grading judgment, or an asymmetry between the two systems that has nothing to do with judgment quality — without requiring a new, comparably-tuned C_LLM prompt to be built and re-evaluated: two independent, zero-additional-cost checks already point the same direction, and it is not in ConceptGrade's favour.

Inference-compute disclosure. Only Layer 2 (KG comparison) is offline, pure graph matching with no LLM involved; Layers 3–5 each make independent LLM calls (cognitive-depth classification: 1; misconception/false-belief detection: 2; verification: 1), and Layer 1's self-consistency extraction adds 3 more via majority-vote. **The evaluated C5_fix configuration therefore issues 7 LLM calls per graded response, versus 1 for C_LLM** – a 7× inference-compute asymmetry that confounds the reported gain, independent of the tuning-budget confound (§VII-B), which also reports a call-budget-matched ablation testing whether this asymmetry explains the gain. It doesn't.

VIII. Conclusion

We set out to treat grading as a knowledge-graph comparison problem, and built ConceptGrade, a five-layer, concept-aware ASAG framework for Computer Science education, to test that idea. On the KG-aligned Mohler et al. (2011) sample ($n = 1{,}262$, 46 questions), it reaches Pearson $r = 0.784$ and QWK$=0.524$, an 8.2% MAE reduction over the identical-model LLM zero-shot baseline. That reduction is significant at the response level but only marginal once responses are grouped by question (§V), and ConceptGrade's Pearson correlation actually comes in *worse* than the baseline's (0.784 vs. 0.790). It's also worth being precise about what's being compared here: a tuned, multi-stage system against an untuned, single-call baseline (§VII-B). Give the baseline the same 7-call budget and C5_fix still comes out ahead at the response level (+4.4% MAE, $p = 0.0053$), but that edge disappears at the question-clustered level ($p = 0.4219$).

Pool across all three datasets and the fixed-effects estimate looks solid, $d_z = -0.105$ ($p < 0.0001$). But that assumes the three datasets share one true effect, and they don't – the more appropriate random-effects model, given substantial between-dataset heterogeneity ($I^2 = 73.2\%$), gives a 95% CI that essentially touches zero ($[-0.169, +0.002]$). We don't treat the pooled result as confirmatory (§V-C). One architectural finding is clearer, though: a real-data 3-condition ablation (§VI) shows the pre-verifier KG-grounded score is far *worse* than the zero-shot baseline (-86.9% MAE). It's the verifier stage, not the knowledge graph, that produces almost all of ConceptGrade's small net gain.

Self-consistency ensembling (K=7, temperature 0.7, §V-E) improves response-level MAE over a *single-call* baseline on Mohler and DigiKlausur. But when we checked it against an equally-resourced C_LLM×7 baseline instead, the story got more complicated: the response-level MAE gap holds (+7.0%/+6.4%, $p < 0.001$), yet question-clustered significance and LOOCV robustness mostly don't survive the fairer comparison (Mohler collapses to $0/46$ folds; DigiKlausur's correlation advantage even reverses). Kaggle ASAG benefits under neither comparison, which fits its concept-extraction failure. The honest version of this claim is narrower than the first one we would have written: self-consistency gives a real, precisely-estimated response-level improvement over a fairly-resourced baseline on two datasets, at 7× inference cost, but its question-level robustness only holds up against a single-call baseline, not the fair control reported here.

Beyond raw accuracy, ConceptGrade also produces structured diagnostic feedback: matched and missing concepts, incorrect relationships, Bloom's/SOLO levels, and misconception tags – though the machine-IRR check on those tags only reaches $\kappa_{\text{micro}} = 0.116$ (slight agreement on real data), while human IRR on the grading task itself comes in much higher ($r = 0.7833$/QWK$=0.7986$, §VII-B). None of this needs task-specific fine-tuning beyond building the knowledge graph.

Future work includes: (1) testing on other CS topics via the Programming/OOP knowledge graph; (2) human-annotated Bloom's and SOLO ground truth to validate the classifiers directly; (3) integrating with a learning management system for live classroom use; (4) building the knowledge graph and misconception taxonomy with a third party (e.g. SIGCSE crowdsourcing) to remove single-author bias; and (5) a multi-dataset meta-analysis that includes frontier-LLM baselines.

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